conductivity passes through a maximum. FAB-MS shows the presence of the chloride coordinated to both one and two glycol molecules and presumably the composition of the two species will depend upon the composition of the two components in the mixture.

It has previously been shown that for other DESs such as those based on urea or acetamide, the mobile species correspond to the chloride anion bound to two hydrogen bond donors.[7] The results shown above suggest the high conductivities and low viscosities presented in Figures 3.2 and 3.3 arise from smaller mobile species which results from weaker anion-HBD interactions.

***Table 3.1**: Energies of activation of both viscosity and conductivity for 1,2-ethanediol based systems*

Mol. % ChCl	E_η (kJmol^{-1})	ln η_0	E_Λ (kJmol^{-1})	ln κ_0
5	42.28	-14.67	-17.65	8.60
10	39.05	-13.22	-12.82	7.01
15	37.71	-12.56	-14.45	7.91
20	39.26	-13.10	-11.95	6.90
25	35.48	-11.19	-13.37	7.46
30	79.50	-29.15	-12.73	7.11
33	84.48	-31.15	-12.02	6.79

As more of the salt is added to the system it would be expected that the density of the system should increase at a linear rate. This can be seen in Figure 3.8. There is however only a slight change in the density. The addition of 33% molar ratio of ChCl to 1,2-ethanediol results in a small increase of 0.007 gcm^{-3}. This results in an increase of only 0.6% of the density of the pure liquid. This would suggest that there is very little change in the strucutre of the system.

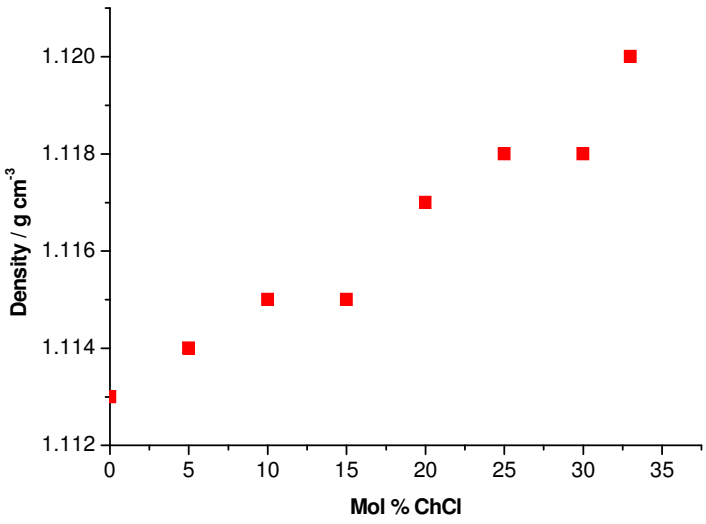


***Figure 3.8**: Plot of molar percentage composition of ChCl versus density at 293K for the 1,2-ethanediol systems*

Using the density data the molar volume V_m can be calculated using

$$V_m = M_r / \rho \quad \textbf{(3.11)}$$

Where M_r is the relative molar mass of the liquid calculated from

$$M_r = x_{ChCl} M_{ChCl} + x_{HBD} M_{HBD} \quad \textbf{(3.12)}$$

where x is the mole fraction and M the relative molar mass of the two components. The actual volume occupied by the components can be calculated in a similar manner i.e.

$$V_{comp} = (x_{ChCl} V_{ChCl} + x_{HBD} V_{HBD}) N_A \quad \textbf{(3.13)}$$

Where V is the molecular volume of the component and N_A the Avogadro constant. Using this approach the fractional molar free volume can be obtained using

$$V_{free} = (V_m - V_{comp}) / V_m \quad \textbf{(3.14)}$$

A liner decrease in V_{free} with mole fraction ChCl can be observed in Figure 3.9 representing an almost 3 % contraction of the liquid compared to pure 1,2-ethanediol.

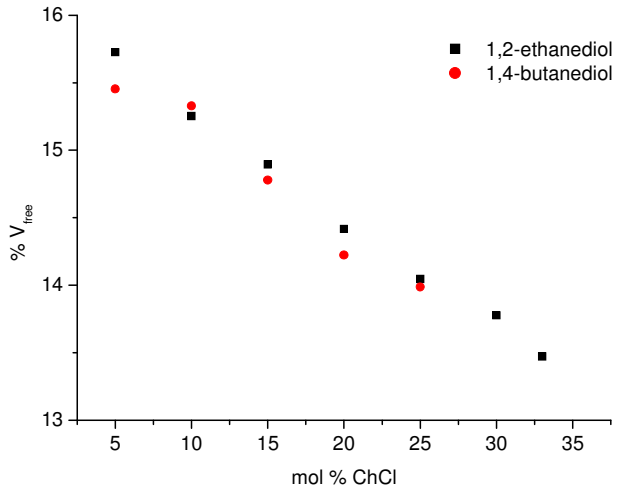


***Figure 3.9**: Plot of V_{free} as a function of mol % ChCl*

3.1.2 1,4-Butanediol and 1,6-Hexanediol

Mixtures of ChCl with 1,4-butanediol (6) were made up ranging from 5% ChCl up to 25% ChCl. It was not possible to go to a higher concentration of ChCl than this as the mixture began to phase separate. The viscosity, conductivity, surface tension and density were all measured and the results are presented below. The results are exactly analogous to those observed for 1,2-ethanediol shown above.

It should be noted that 1,6-hexanediol is a solid at room temperature (melting point 40-43 °C) and so no density data could be obtained due to the high temperature required to form a liquid. The phase diagram of 1,6-hexanediol with ChCl is shown in Figure 3.10.

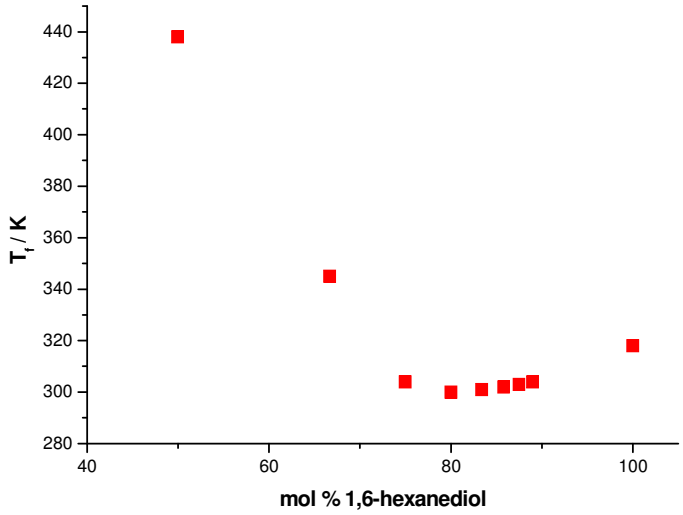
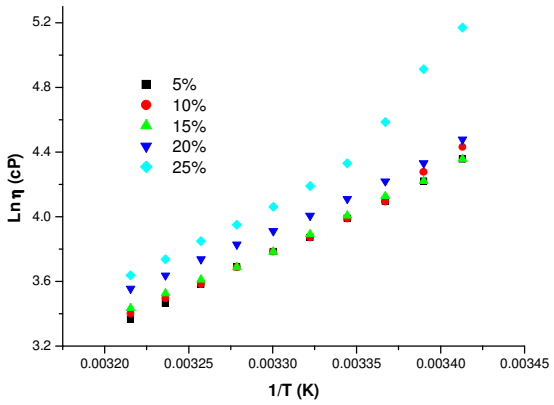
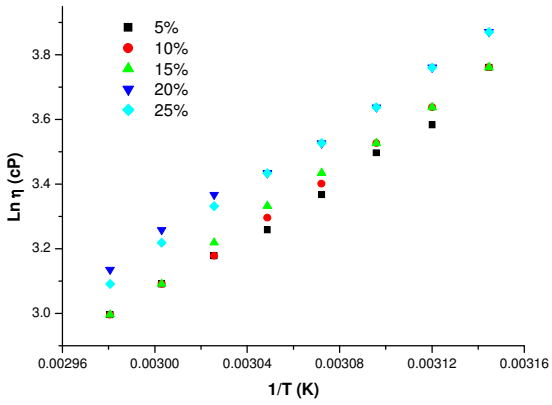
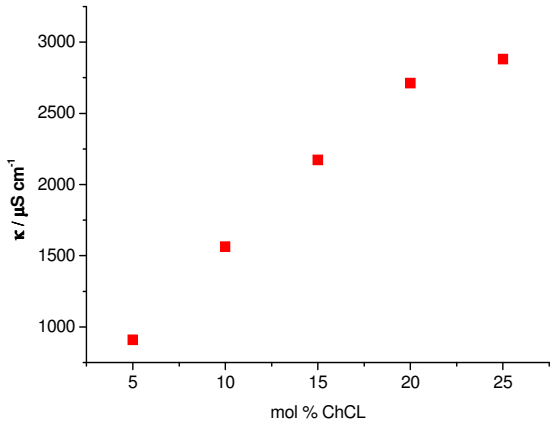
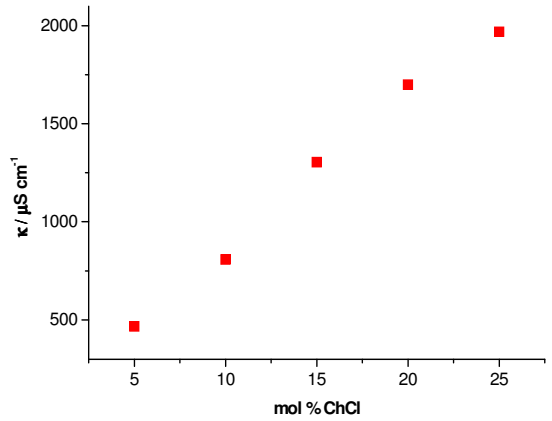
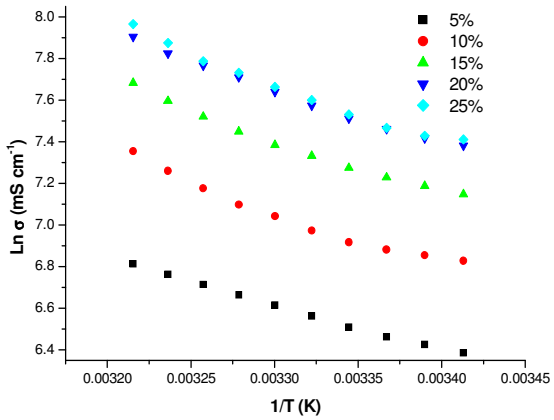
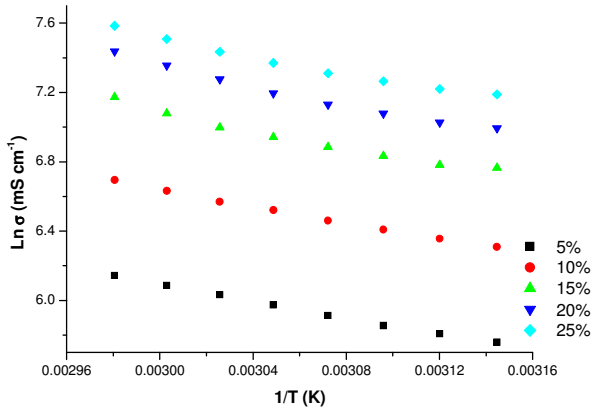


***Figure 3.10**: Phase diagram of 1,6-hexanediol with ChCl*

The lowest freezing point appears to be at a ratio of 4:1 1,6-hexanediol: ChCl. The depression is very small compared to the other DESs studied previously such as malonic acid where the depression of freezing point is 212 ^{o}C. The depression of freezing point compared to an ideal mixture of the two components at the eutectic composition is 69 ^{o}C this corresponds to a mass fraction of HBD of 67% and fits relatively well to the empirical model presented previously for DESs.[1-4,6-9,17]

Figures 3.11 and 3.12 show that the viscosities of the C_4 and C_6 diols are significantly larger than those of the C_2 diol. A concomitant decrease in the conductivities of the larger diols is also observed. The viscosity of these systems shows some strange behaviour. The viscosity of 5%, 10% and 15% is almost the same and there is a significant change in viscosity for 20% and 25% mixtures which are similar to each other.

As with the 1,2-ethanediol the conductivity of the 1,4-butanediol and 1,6-hexanediol systems increased as more ChCl was added, which is what would be expected since there are more charge carrying species in the system. Figures 3.13 and 3.14 however show no decrease in the conductivity at high ChCl concentrations as shown in Figure 3.4, but this is probably because such large concentrations could not be obtained with the higher diols. The energies of activation of conductivity and viscosity were calculated for each of the systems and the results are shown in Table 3.2.

1,4-Butanediol	1,6-Hexanediol
	
***Figure 3.11**: Plot of T^{-1} vs. log η as a function of ChCl*	***Figure 3.12**: Plot of T^{-1} vs. log η as a function of ChCl*
	
***Figure 3.13**: Plot of mol % ChCl vs. conductivity at 311K*	***Figure 3.14**: Plot of mol % ChCl vs. conductivity at 335K*
	
***Figure 3.15**: Plot of T^{-1} vs. log κ as a function of ChCl composition*	***Figure 3.16:** Plot of T^{-1} vs. log κ as a function of ChCl composition*

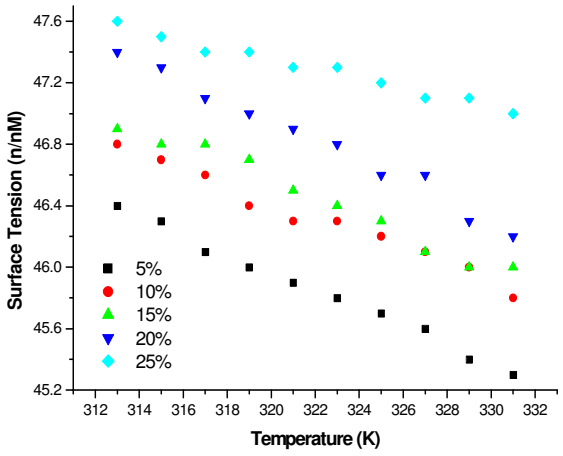


Figure 3.17: *Plot of T vs. γ as a function of ChCl composition*

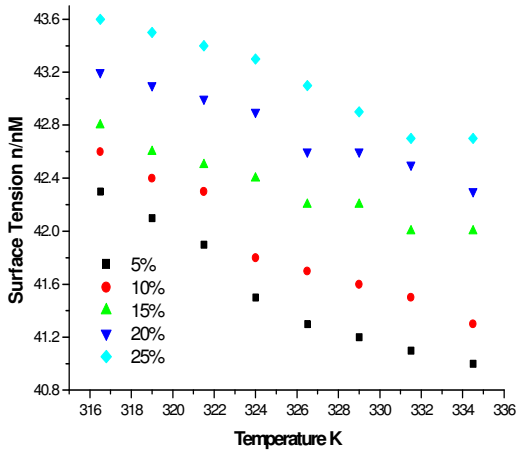


Figure 3.18: *Plot of T vs. γ as a function of ChCl composition*

Table 3.2*: Energies of activation of both viscosity and conductivity for 1,4-butanediol and 1,6-hexanediol systems*

1,4-butanediol				
Mol. % ChCl	**E_η (kJmol^{-1})**	**log η_0**	**E_Λ (kJmol^{-1})**	**log κ_0**
5	40.71	-12.37	-18.36	13.91
10	42.01	-12.88	-22.11	15.85
15	38.61	-11.52	-22.24	16.24
20	37.92	-11.13	-22.35	16.52
25	62.13	-20.51	-23.80	17.13
1,6-hexanediol				
5	37.58	-10.50	-19.78	13.23
10	39.06	-11.02	-19.54	13.69
15	38.50	-10.80	-20.61	14.52
20	36.21	-9.83	-22.81	15.59
25	38.78	-10.80	-20.14	14.78

Table 3.2 shows that the $\mathbf{E_\eta}$ values are comparable with those observed for 1,2-ethanediol at the same mole fraction of ChCl (35-40 kJmol^{-1}). In the 1,4-butanediol and 1,6-hexanediol systems the average hole sizes calculated from the surface tension data were 1.55 and 1.68 Å respectively. Analogous to the results above for 1,2-ethanediol it is most likely that the mobile species are either 'free' chloride ions or those bound to one glycol molecule.

The density of the systems is expected to increase as more quaternary ammonium salt is added to the system and this is what was observed in Figure 3.19. Although the

density changes observed are significantly larger than those shown in Figure 3.8 they equate to exactly the same change in molar free volume (Figure 3.9) suggesting that the same structuring occurs in both liquids.

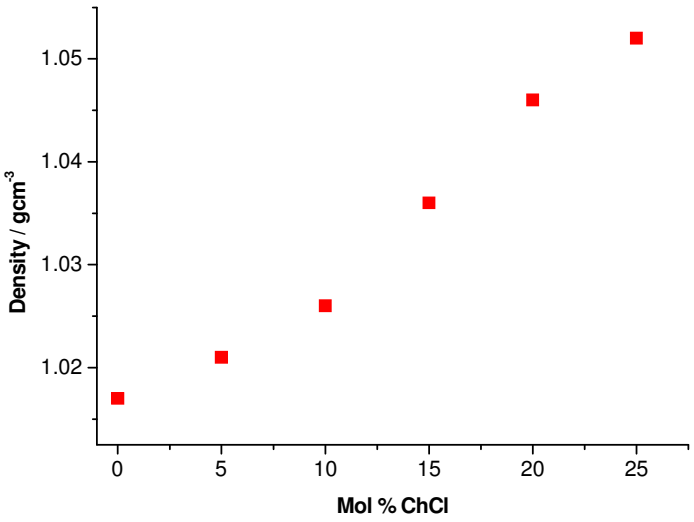


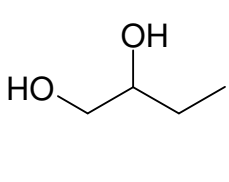
***Figure 3.19**: Plot of molar percentage composition of ChCl versus density at 293K for the various 1,4-butanediol systems*

This section has shown that the chain length of the diol significantly affects the conductivity and viscosity of the mixtures. It has been shown that the cause of the instability in the glycol based eutectics arises from the low freezing point to the hydrogen bond donor. Since the depression of freezing point is related to the strength of interaction between the chloride anion and the hydrogen bond donor; a non-acidic proton combined with a low freezing point of the glycol means that a very shallow eutectic is formed. This may also explain why aliphatic alcohols do not form eutectics with ChCl.

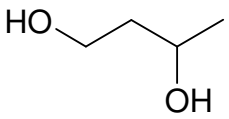
3.2 Effects Hydrogen Bond Donor Group Positioning

The effects of altering the length of the glycol chain has been investigated in Section 3.1, but the effect of the positioning of the alcohol groups along the alkane chain has not been looked at. The physical properties of the various isomers of butanediol were investigated.

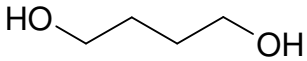
This section looks at the physical properties (viscosity, conductivity, surface tension and density) of the various isomers of butanediol (1,2-, 1,3-, 1,4- and 2,3-butanediol) when mixed with ChCl, similar to the experiments carried out in **Section 3.1.**



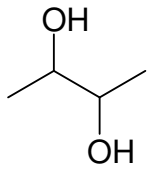
1,2-butanediol (11)



1,3-butanediol (12)



1,4-butanediol (6)



2,3-butanediol (13)

This section is looking at what affect the organic groups on the hydrogen bond donor molecules have. The interaction between the OH moiety and the chloride anion are unknown (current EXAFS studies are attempting to quantify these) however one way in which the different isomers could affect the packing density in the liquid is shown schematically in Figure 3.20.

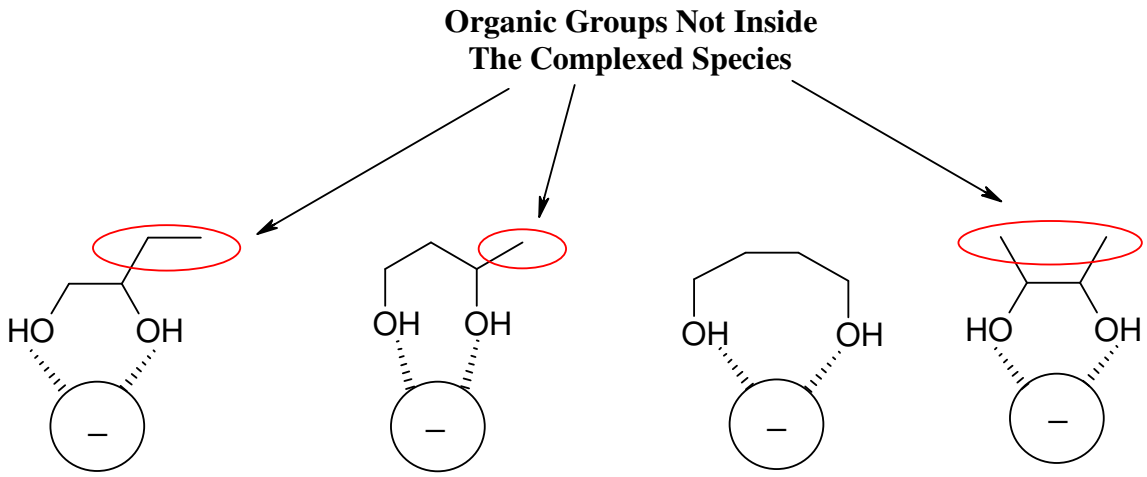


***Figure 3.20**: Schematic representation of possible complex species that could be formed between the various isomers of butanediol with the chloride anion*

One way of gaining an insight into the structural differences occurring in the liquids is to analyse the physical properties.

The viscosity of the systems all showed an exponential decrease as temperature is increased. The various plots can be seen in Appendix Figures 7.3, 7.7, 7.11 and 7.15.

It is also seen that by adding ChCl to the system there is an increase in the viscosity of the liquid, as was seen in the case of the glycol based systems. Figure 3.21. shows that the viscosity decreases in the order the order of 2,3 > 1,3 > 1,4 > 1,2.

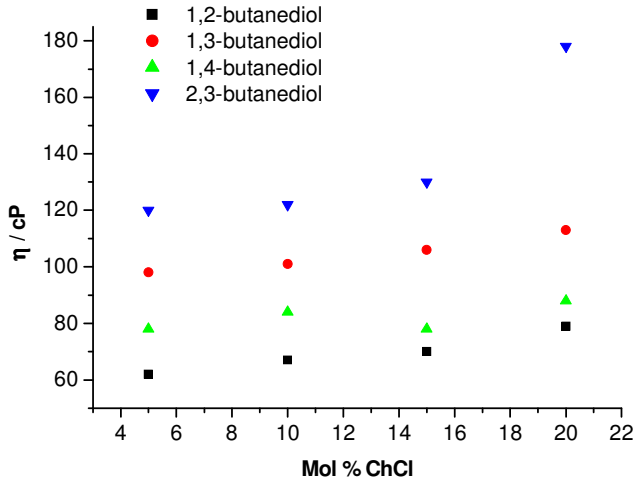


***Figure 3.21**: Viscosity of the various isomers of butanediol as a function of molar composition of the system in terms of ChCl at 298K.*

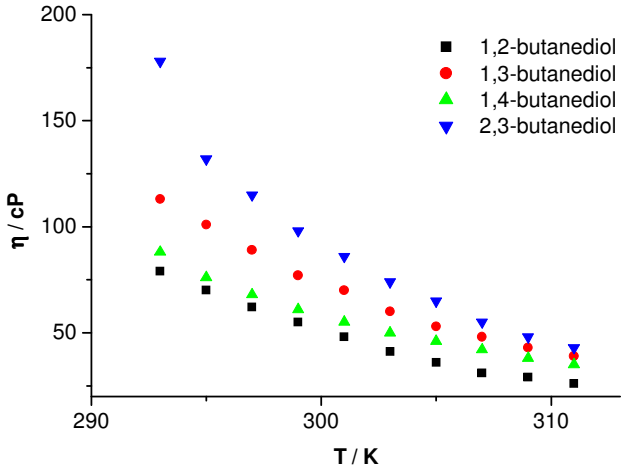


***Figure 3.22**: Plot of viscosity as a function of temperature for all the butanediol systems at 20% ChCl*

Figure 3.22 shows the change in viscosity with temperature and it can be seen that the position of the hydrogen bond donors can have a large effect on the physical properties of the system. Simply by shifting the hydrogen bond donors one carbon over from a 1,2 to a 2,3 position, the viscosity more than doubles. A plot of inverse

temperature versus log viscosity for all systems at 20% ChCl can be seen in Figure 3.23. This plot has been used to calculate the energy of activation of viscosity (E_η) which can be seen in Table 3.3. Figure 3.20 shows a very linear correlation in each case.

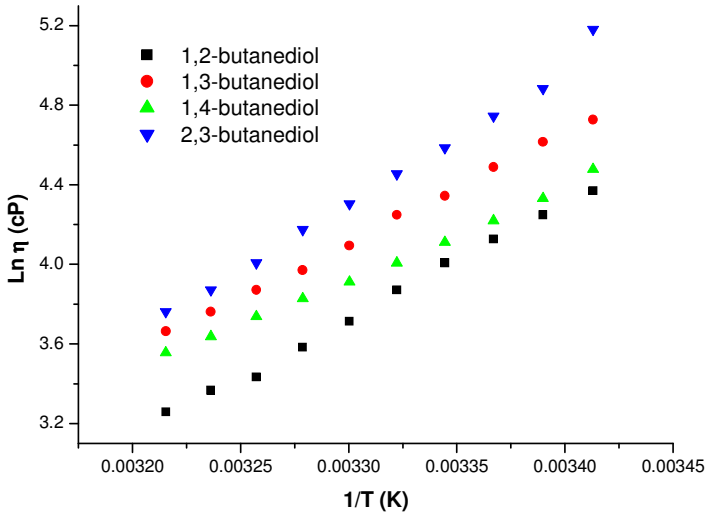


Figure 3.23: *Plot of inverse temperature versus log viscosity for all the butanediol systems at 20% ChCl*

Figure 3.24 shows the conductivity of the various butanediol isomers as a function of ChCl composition

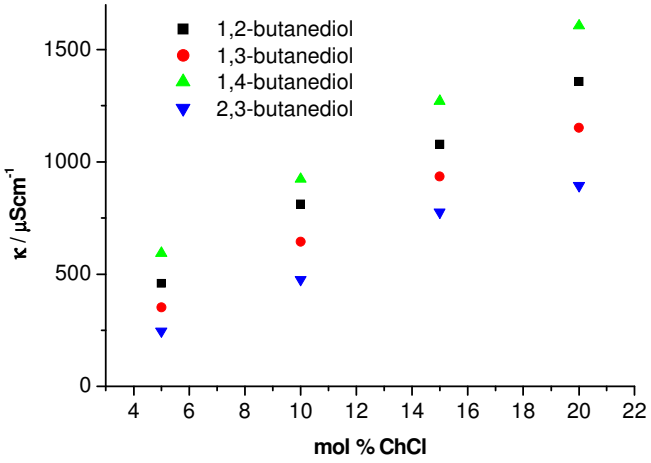


Figure 3.24*: Conductivity of the various isomers of butanediol as a function of molar composition of ChCl*

From what has been seen in the viscosity experiments, it would be expected that the more viscous systems, in this case the 2,3 system, should be the least conductive, especially since each isomeric system contains exactly the same amount of charge

carrying species. Hence the least viscous systems should conduct more. As can be seen in Figure 3.25, this is not the case.

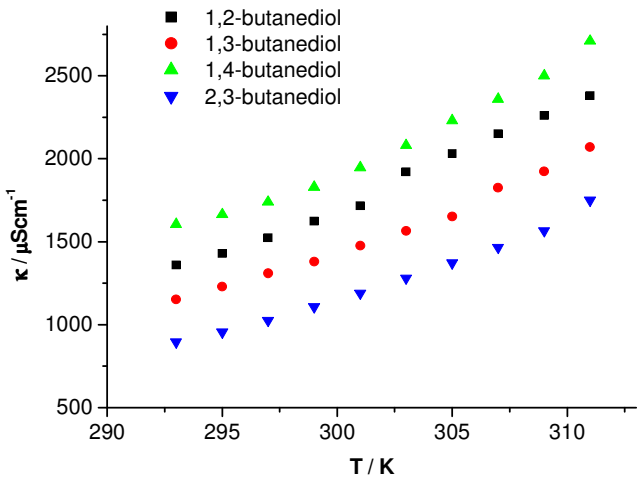


***Figure 3.25**: Plot of conductivity as a function of temperature for all the butanediol systems at 20% ChCl*

From the viscosity results the proposed trend for conductivity would be 2,3 < 1,3 < 1,4 < 1,2. However the order of the 1,4 and 1,2 isomers is in fact inverted suggesting that there is a difference in the way 1,4-butanediol complexes to the chloride ion of ChCl compared to the other butanediols. A possible explanation could be that when 1,4-butanediol complexes to the chloride ion, it is able to "wrap" around the anion more tightly than the other diols due to the hydrogen bond donors being on the end of the alkane chain, which would possibly result in a smaller volume. The smaller the complex, the more easily it should be to move around in the system and pass the electrical charge between the other charge carriers. An alternative explanation may come from the fact that 1,4-butandeiol may only complex one molecule to the anion, rather than two as is believed to be the case with the other butanediols.

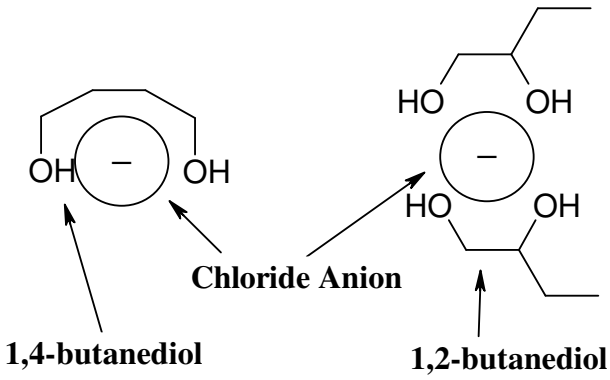


***Figure 3.26**: Diagrammatic representation of the possible complexes of the various isomers of butanediol*

From the diagram shown in Figure 3.26, it is possible that one molecule of 1,4-butanediol has a large enough space between the hydrogen bond donor groups, that it envelopes the chloride anion so much that only one molecule can complex around it. It is thought that the space between the hydrogen bond donor groups on the other butanediols, for example 1,2-butanediol, is not large enough to completely "wrap" around the chloride anion and so more than one molecule can complex.

A plot of inverse temperature versus log viscosity for all systems at 20% ChCl can be seen in Figure 3.27. This plot has been used to calculate the energy of activation of conductivity (E_Λ) which can be seen in Table 3.3.

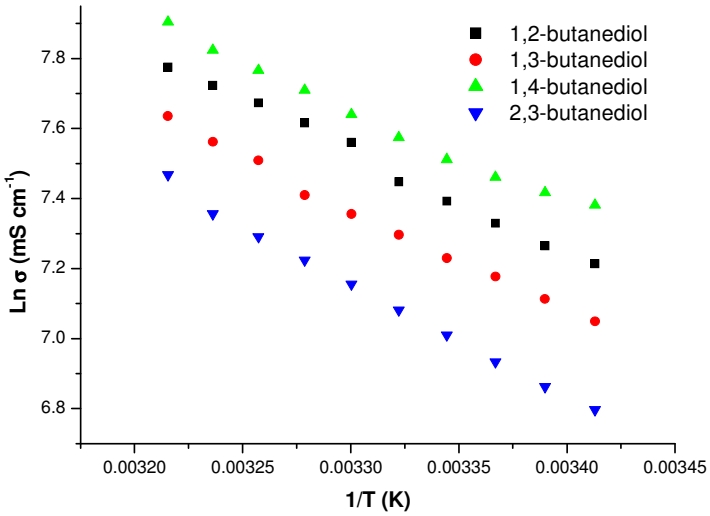


***Figure 3.27**: Plot of inverse temperature versus log viscosity for all the butanediol systems at 20% ChCl*

As has been suggested throughout Section 3.1., as the viscosity of the system decreases, the surface tension of the system should be seen to increase as well. The surface tension data for the four isomers can be seen in Figure 3.28.

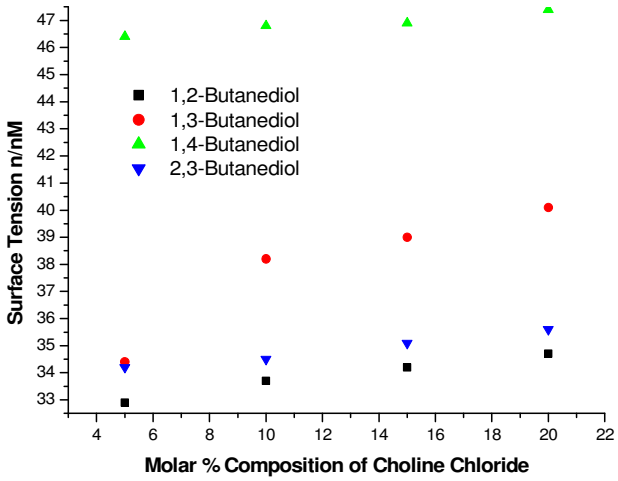


Figure 3.28: *Surface tension of the various isomers of butanediol as a function of molar composition of the system in terms of ChCl*

The data shown in Figure 3.28 shows a different trend to that obsereved for the viscosity results. The results for viscosity followed the trend 2,3 > 1,3 > 1,4 > 1,2, whereas the surface tension trend follows 1,4 > 1,3 >, 2,3 > 1,2. 1,4-butanediol has the highest surface tension, due to the nature of its intermolecular hydrogen bonding, since 1,4-butanediol has hydroxyl groups on the two ends of the molecule, the hydrogen bonding causes a linear "chain" which essentially holds the dialcohol in a fixed conformation which therefore requires greater force to penetrate through then the other butanediol isomers. The other diols, when complexed, have the hydrogen bond donors locked into a fixed position but the substituents on the outside of the hydrogen bond donors are not locked into any conformation and so have free rotation. This results in less dense packing of the system, which gives the systems a lower surface tension.

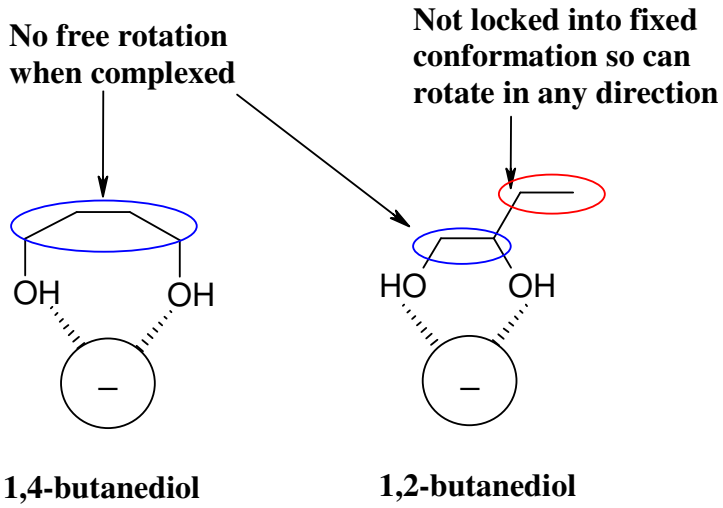


***Figure 3.29**: Diagrammatic representation of the possible free rotations of the various butanediol isomers*

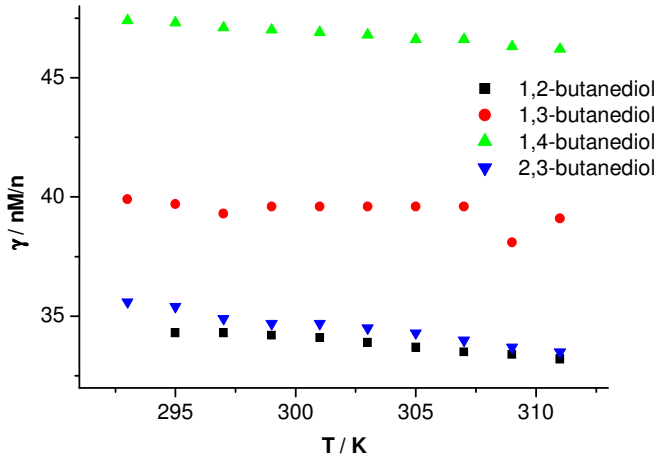


***Figure 3.30**: Plot of surface tension as a function of temperature for all the butanediol systems at 20% ChCl*

The plot in Figure 3.30 shows that there is quite a variation in the surface tension of the various isomeric systems. Again by using the surface tension data, the average hole sizes have been calculated and compared to the energies of activation of viscosity and conductivity that have also been calculated. These data can be seen in Table 3.3. Using Figure 3.7 which shows the correlation between E_η and r_i/r_h it can be seen that for the 1.4- isomer the average ion size is 2.31 Å whereas for the 2,3- isomer the average ion size is 3.03 Å.

***Table 3.3**: Select energies of activation of both viscosity and conductivity for all butanediol isomer based systems*

HBD	Mol % ChCl	Eη ($kJmol^{-1}$)	ln η0	EΛ ($kJmol^{-1}$)	ln κ0	rh (Å)
1,2-butanediol	20	48.38	-15.48	-24.72	17.35	1.802
1,3-butanediol	20	45.74	-14.05	-24.48	17.09	1.676
1,4-butanediol	20	37.92	-11.13	-22.35	16.52	1.542
2,3-butanediol	20	57.31	-18.44	-27.53	18.09	1.779

More insight can be gained from the density of each of the systems. Figure 3.31 shows that as expected the density increases with increasing ChCl concentration.

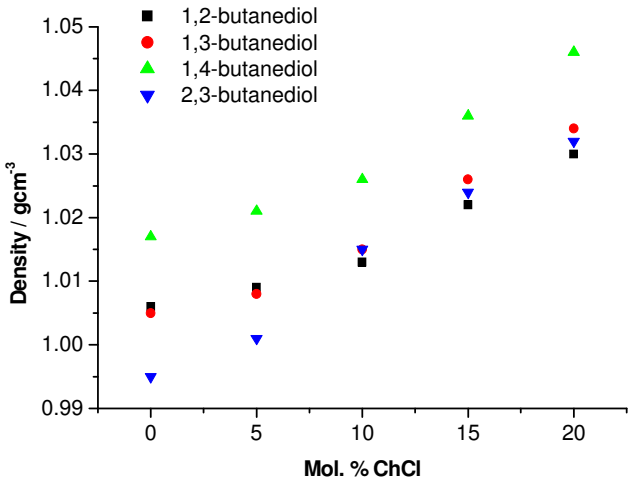
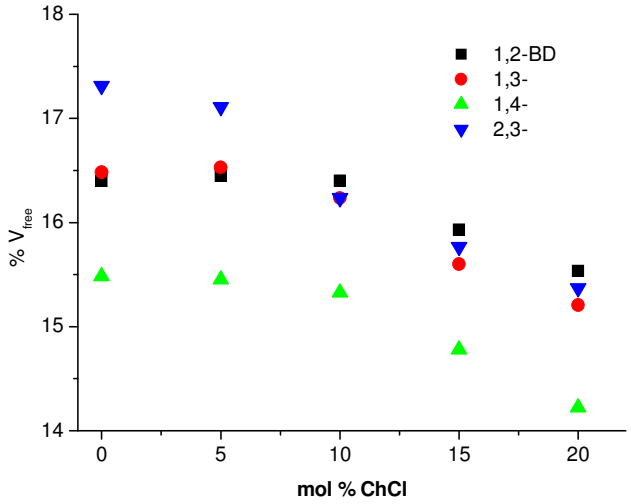


***Figure 3.31**: Density of the various isomers of butanediol as a function of molar composition of the system in terms of ChCl*



l

***Figure 3.32**: Percentage molar free volume of the various isomers of butanediol as a function of molar ChCl composition*

At low ChCl content the salt has little effect upon the free volume of the liquid. Above 10 mol % a linear decrease in V_{free} is observed. It is noteworthy that the 1,2-, 1,3- and 2,3- isomers have similar V_{free} values whereas the 1,4- isomer has significantly less free volume suggesting a more ordered structure. This explains the higher viscosity and surface tension of the 1,4- isomer compared to the other isomers, but not why the 1,4, isomer has a significantly larger conductivity than the other liquids. This difference must come from a change in the mobile species as discussed above. The data presented here do not, however, allow a definitive mobile species to be identified.

3.3 Variation of the Number of Hydrogen Bond Donating Groups

So far this report has only investigated the effects of using two alcohol groups on each hydrogen bond donor when forming a DES with ChCl. This section will look at the effect of taking the basic 1,2-ethanediol structure and adding $(CHOH)_x$. The compound investigated is glycerol.

3.3.1 Physical Properties of Glycerol with ChCl

With glycerol, viscosity, conductivity and surface tension have all been measured as functions of temperature and molar percentage composition with ChCl.

As with all viscosity measurements carried out during this study, the results for glycerol show exponential decay as the temperature is increased. Pure glycerol is a viscous liquid. In fact it is much more viscous 1,2-ethanediol (approximately 20 times greater) and so initially results are expected to show very high viscosities when ChCl is added. Figure 3.31 however shows that as the ChCl content increases the viscosity actually decreases.

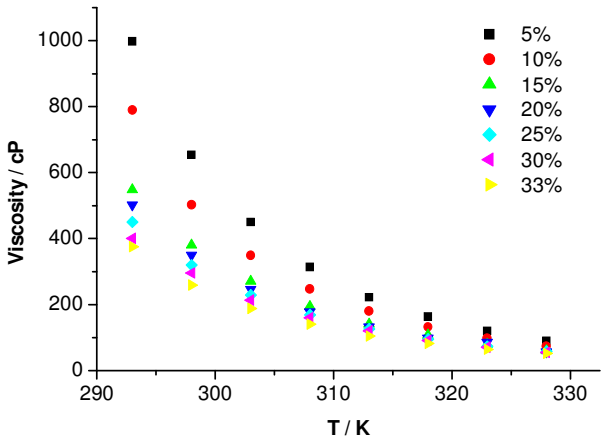


***Figure 3.33**: Plot of temperature versus viscosity as a function of ChCl composition for glycerol the systems*

This trend is more obvious when the viscosity is taken at one temperature and compared as a function of molar percentage composition of ChCl. This can be seen in Figure 3.34.

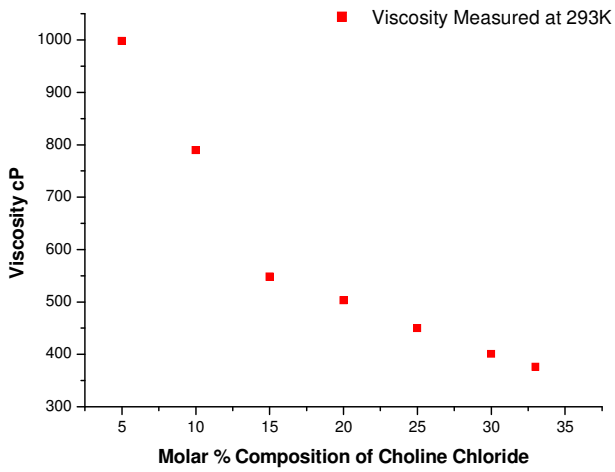


***Figure 3.34**: Plot of molar percentage composition of ChCl versus viscosity at 293K for the glycerol systems*

The reason for this is thought to be due to the additional alcohol group increasing the amount of intermolecular forces in the pure system which results in a 3-D as shown in Figure 3.35.

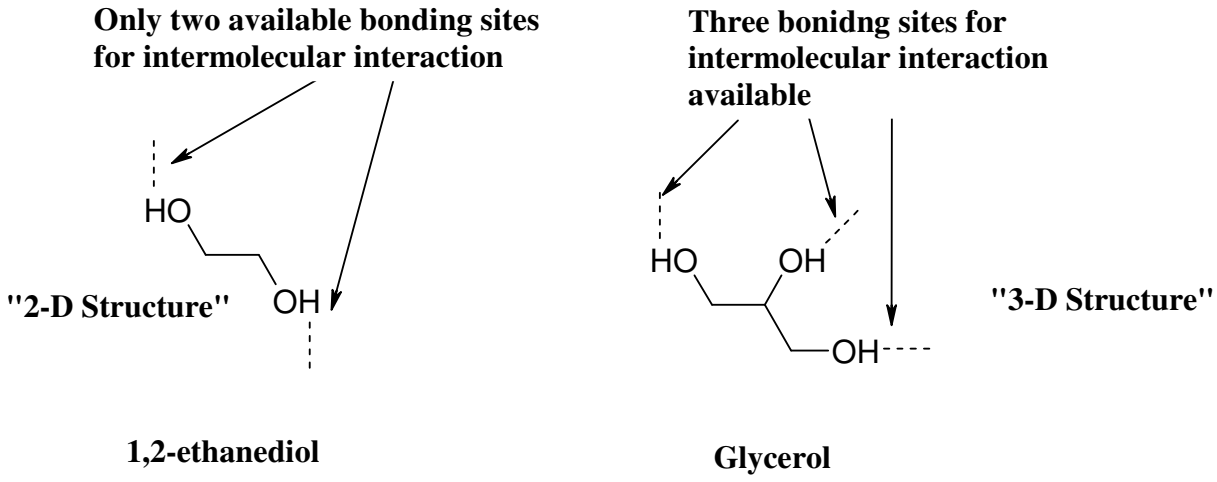


***Figure 3.35**: Schematic diagram of the interactions occurring in 1,2-ethanediol and glycerol*

The fact there are three hydrogen bond donor sites in glycerol as opposed to two in 1,2-ethanediol, means that there is a greater degree of hydrogen bonding to other molecules and so the addition of ChCl interrupts this structure since some of the alcohol groups no longer interact between each other they interact with the ChCl. The end result of this is a system with a greater degree of freedom, hence the reduction in viscosity.

With viscosity decreasing as more charge carrying species are added to the system, it is largely expected that conductivity should increase as more ChCl is added. Despite the viscosity decreasing as more ChCl is added, the viscosity is still relatively high and so the conductivity would be expected to significantly smaller than for 1,2-ethanediol. This is exactly what is seen and can be seen in Figure 3.36.

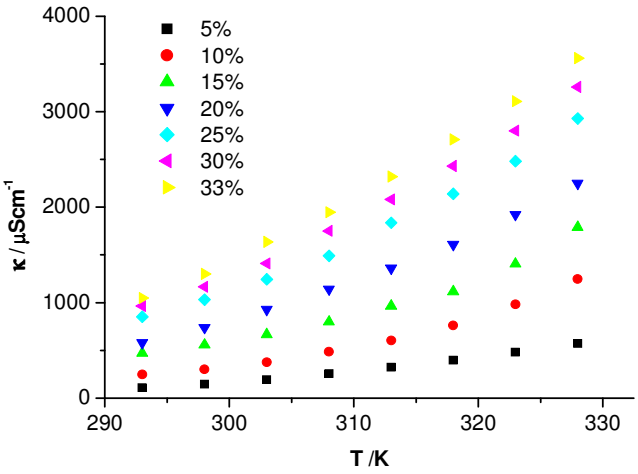


***Figure 3.36**: Plot of temperature versus conductivity as a function of ChCl composition for glycerol systems*

However, what is interesting in these results is the rate of increase of conductivity as more ChCl is added. Since viscosity is decreasing as more charge carrier is added, it would be expected that the rate of increase of conductivity would be more of an exponential growth as molar percentage composition of ChCl increases than a linear relationship. This however is not the case. As can be seen in Figure 3.37, the rate of increase in conductivity is linear.

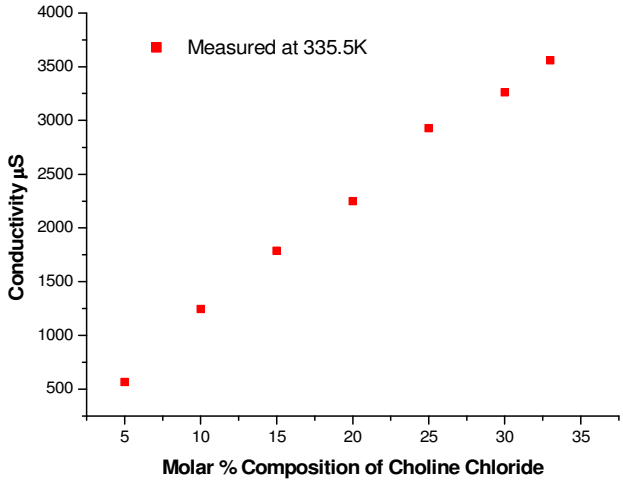


***Figure 3.37**: Plot of molar percentage composition of ChCl versus conductivity at 335.5 K for glycerol systems*

The surface tensions of the various compositions show very linear relationships as function of temperature. As would be expected from the viscosity results, the surface tensions decrease as more ChCl is added to the system. A plot of temperature versus viscosity can be seen in Figure 3.38. This is further evidence that the ChCl is breaking up the intermolecular forces of the glycerol as described in Section 3.3.1. What is interesting to note is that although 1,2-ethanediol and glycerol have significantly different viscosities in the pure state (20 and 1412 cP respectively at 303K) the addition of 33 mol% ChCl results in eutectic liquids with broadly similar viscosities (36 and 376 cP respectively).

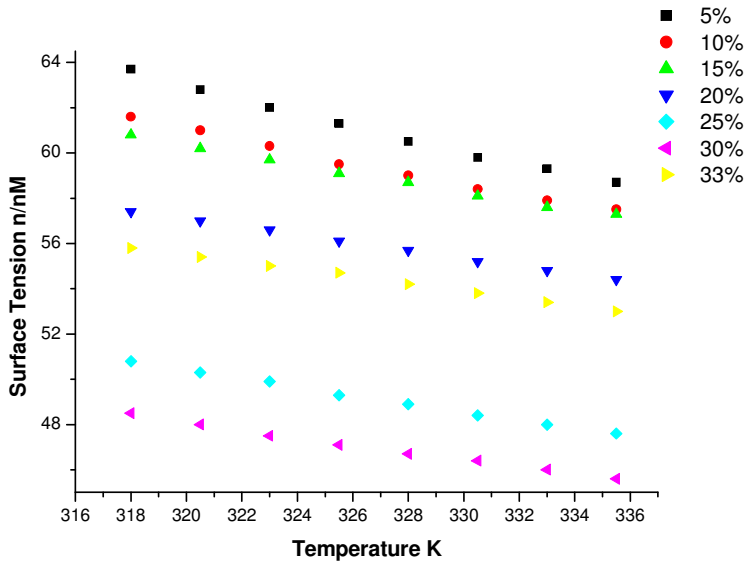


***Figure 3.38**: Plot of temperature versus surface tension for the glycerol systems*

There is however a slight deviation from the predicted results. Although surface tension decreases as more ChCl is added to the system, there is an increase from 30% to 33%, despite the viscosity being at its lowest at 33%. This can be seen in Figure 3.39.

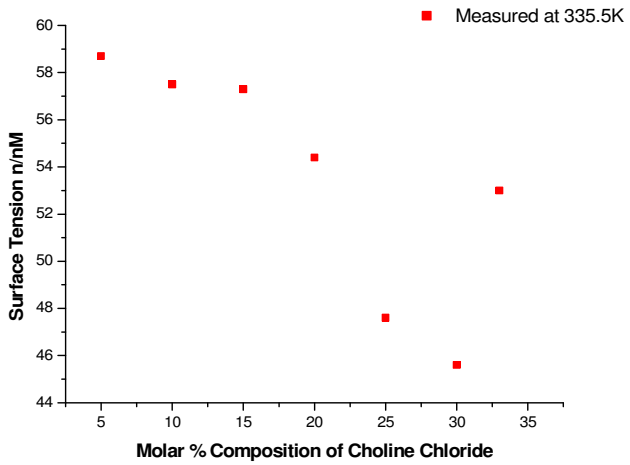


***Figure 3.39**: Plot of molar percentage composition of ChCl versus surface tension for the glycerol systems*

From Figure 3.39, it can be seen that the surface tension of the systems decrease up to 30% ChCl with quite a step drop in surface tension going from 15% to 30%. It can be seen from this plot that from 30% to 33% there is a large rise in the surface tension. This would suggest that there is quite a change in the structure at this composition which equates to a 1:2 ratio of ChCl to glycerol. It is however more likely an indication of some errors in the measurements at 25% and 30%.

From the surface tension data it is possible to calculate the average hole size in the system. This data can be seen in Table 3.4.

***Table 3.4**: Energies of activation of both viscosity and conductivity for the glycerol systems*

Mol % ChCl	E_η / kJmol^{-1}	ln η_0	E_Λ /kJmol^{-1}	ln κ_0	r_h / Å
5	54.72	-15.60	-38.51	20.52	1.386
10	53.35	-15.29	-37.54	20.85	1.409
15	50.24	-14.34	-29.85	18.37	1.418
20	48.26	-13.63	-30.68	18.99	1.458
25	46.99	-13.21	-28.36	18.39	1.551
30	45.46	-12.67	-28.05	18.40	1.588
33	45.06	-12.62	-27.90	18.44	1.481

As a result of the decrease in viscosity it would be expected that the density should decrease as well. This is exactly what is observed and a plot of this data can be seen in Figure 3.40.

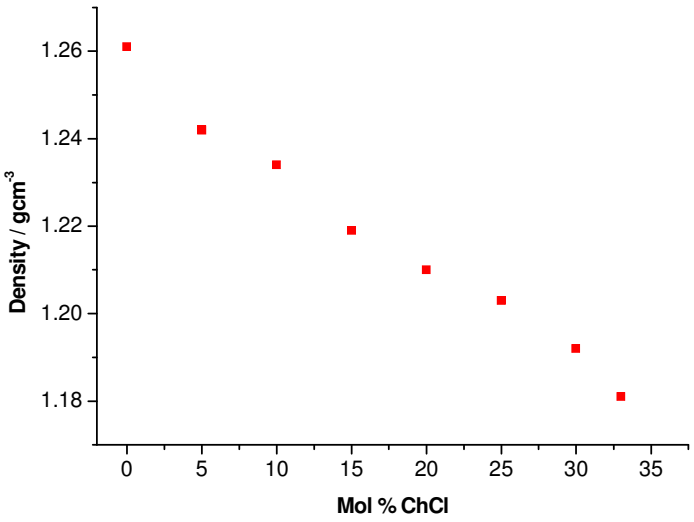


***Figure 3.40**: Density of the glycerol systems as a function of molar composition of the system in terms of ChCl*

Figure 3.9 shows the change in V_{free} for the ChCl/diol systems. The free volume decreases for the ChCl/diol systems but for glycerol it increases as can be seen in Figure 3.41.

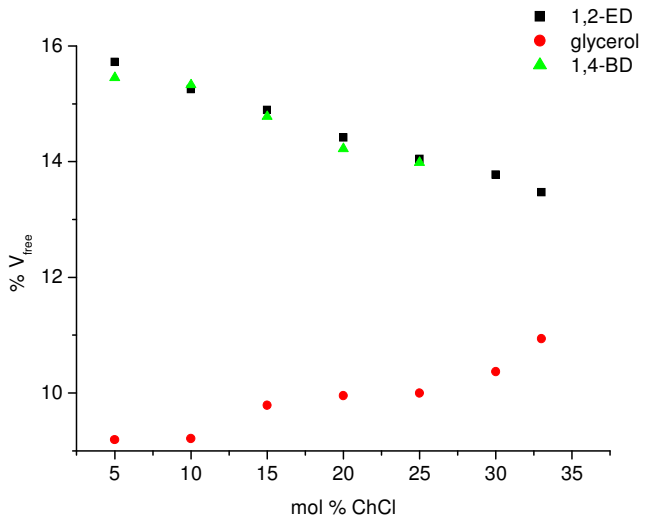


***Figure 3.41**: Plot of V_{free} as a function of mol % ChCl*

This is consistent with the idea that pure glycerol is highly structured and the chloride anion breaks up this structure and increases the free volume of the liquid and decreases the fluid viscosity. It is clear that the structure of the glycerol eutectic is significantly different in structure from that of the diols described above.

3.4 At Which Composition do DESs Become Ionic Liquids?

As described above Walden plots have previously been used to describe the conductivities of ionic liquids. While there is clearly no validity behind the application of the Walden rule to ionic liquids it is interesting to carry out this analysis for DESs to see how they compare with ionic liquids with discrete anions. Figure 3.42 shows a plot of molar conductivity *vs*. fluidity ($1/\eta$) for the data presented in a recent review by Galiński *et al*. for ionic liquids with discrete anions.[29] The scatter in Figure 3.42 is thought to be primarily due to the difference in size between the ions. It has previously been shown that eutectic mixtures of quaternary ammonium salts with hydrogen bond donors form liquids at ambient temperatures and the conductivity of these liquids obey equation (3.9).[14] It has therefore been assumed that DESs can therefore be classified as ionic liquids and Figure 3.42 shows some that these data presented above fit the classification of Angell.[15]

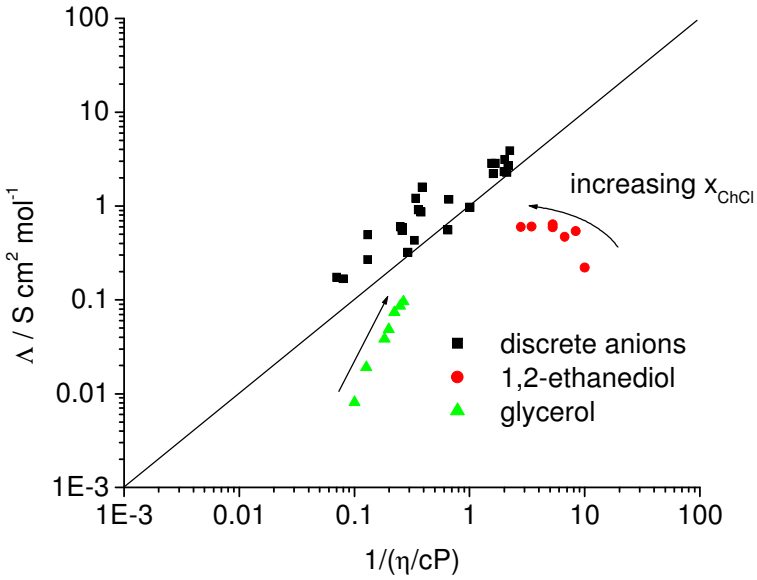


Figure 3.42: *Plot of molar conductivity vs. fluidity for a range of ionic liquids with discrete anions and mixtures of ChCl with glycerol and ethylene glycol at 293 K*

The ability to vary the composition of Lewis or Brønsted acid adds an additional dimension to the tuneability of the eutectic-based ionic liquids. To test the hypothesis that an ionic liquid can become a solution is relatively difficult as most complexing agents are solid at ambient temperature and hence a true test should be conducted at the melting point of the complexing agent. This adds the additional complexity that the interaction between the anion and the complexing agent will be significantly affected by the temperature. Lecocq *et al*. used ^{13}C and ^{35}Cl NMR spectroscopy to show that at 110°C 1BMMICl:1$ZnCl_2$ varies its structure with time from $[BMMI]^+[ZnCl_3]^-$ to $[BMMI…Cl…ZnCl_2]$.[30] What is important however is the behaviour of ionic liquids at ambient temperature. The only exceptions that we have found are with diols and triols which are liquid at ambient temperature and form relatively shallow eutectics when mixed with quaternary ammonium salts.

Figure 3.43 shows the molar conductivity of ChCl in a number of glycolic liquids. The molar conductivity was calculated by

$$\Lambda = \kappa M/\rho \qquad (3.15)$$

where κ is the conductivity, ρ is the density and M is the molar mass which is take as being the molar mass of the two components multiplied by their mole fraction in the mixture. The ionic radii were calculated using commercially available software assuming that each chloride ion is coordinated by one glycol molecule.[28]

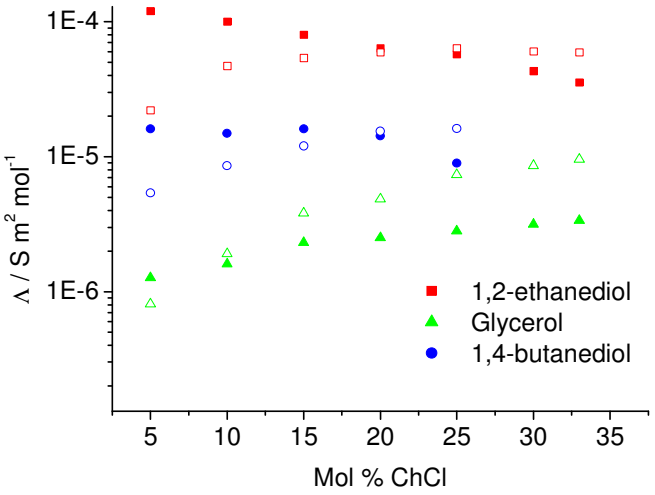


Figure 3.43*: Plot of molar conductivity vs. ChCl content for a variety of mixtures with different glycols at 293 K. Open symbols are the measured values and filled symbols are the values calculated using* ***equation 3.8****.*

At low concentrations most of the ions exist as ion pairs and the molar conductivity is relatively low. As the concentration of salt increases, concentration of triple ions increases which results in a rise in the molar conductivity.

$$3ChCl \leftrightarrow Ch_2Cl^+ + ChCl_2^- \quad \textbf{(3.16)}$$

This is well known to occur in non-aqueous solutions.[20,21] As the salt concentration increases the amount of free solvent decreases and at some point there must be no free solvent molecules as they are all coordinated to the ions. At this point the movement of free "solvent" is prohibited and it is only ionic species that can move. Under these conditions the mobility of the complex species is severely hindered due to their size and the decreased free volume of the liquid. For 1,2-ethanediol a maximum is observed in molar conductivity at approximately 20 mol % ChCl. According to equation (3.9) the limiting molar conductivity will be inversely proportional to the viscosity and it would be expected that the molar conductivity would decrease as hole mobility replaced ion mobility as the dominant factor in charge transport. It is at this composition where hole mobility overtakes ion mobility as the dominant mechanism for charge transfer that we propose an ionic liquid is formed.

Figure 3.41 shows the molar conductivity of the glycolic mixtures as a function of fluidity. It is clear that the molar conductivity values of the dilute solutions differ significantly from the Walden rule which is to be expected as the conductivity is dependent upon the number of charge carriers and the main mobile species which govern the fluidity of the liquid are the glycol molecules. Hence the conductivity is greater than that predicted by equation (3.8). As the concentration of salt increases the number of free solvent molecules decrease and the conductivity approaches that of the other liquids where mobility is limited by hole mobility. Hence the definition by Angell that all ionic liquids obey the same conductivity *vs.* fluidity correlation is a good approximation to determining whether a material is an ionic liquid. A more rigid definition is possibly that an ionic liquid is a fluid where equation (3.8) is valid the justification being that in an ionic liquid the transport of charge is limited by hole mobility.

To determine the composition at which the eutectics in Figure 3.43 change from being ionic liquids to solutions of salt in glycols, the molar conductivity was calculated

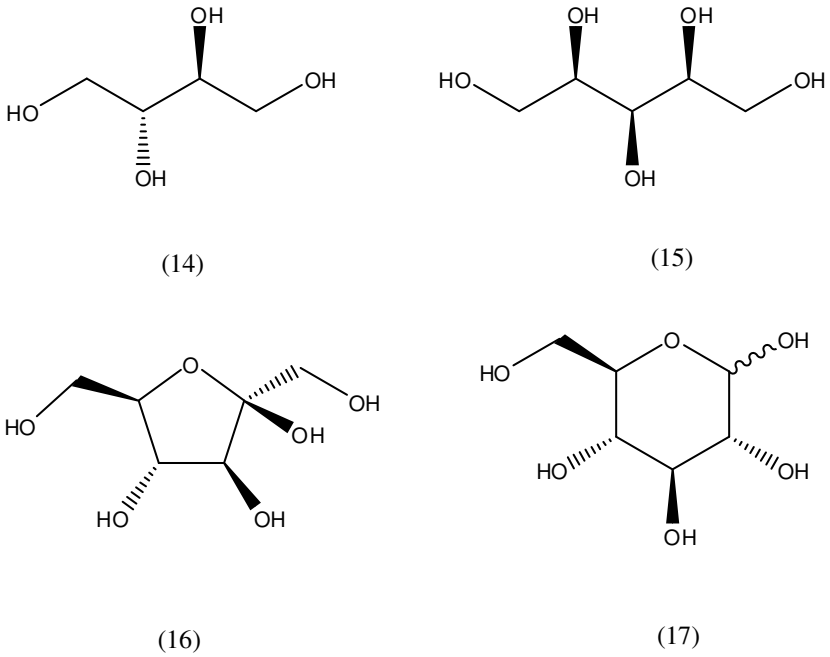
using equation (3.8) and plotted against the measured conductivity and it can be seen that significant deviations are seen up to a composition of 20% 1,2-ethanediol and 1,4-butanediol. Thereupon relatively good correlation was observed. 1,4-butanediol could not be studied above 25 mol % because the temperature at which a homogeneous solution formed was above 20°C. The other isomers of butanediol (1,2-, 1,3- and 2,3-) were also tested and showed similar trends to those observed for 1,4-butanediol but these all reached supersaturation above 20 mol% at 20°C and hence approached the properties of ionic liquids but could not exhibit the decrease in molar conductivity with concentration.

For glycerol the correlation between the values calculated from equation (3.8) and the measured values is closer a lower concentrations. This could be seen as an exception that shows the validity of the assumption, in so far as the very high viscosity of the pure fluid and the low free volume, as indicated by the high surface tension, mean that charge transport is still limited by hole transport, only in this exceptional case the mobility of the "solvent" molecules (i.e. glycerol) is not due to them all being complexed to the Cl^- but rather in this case they are hindered from moving by other solvent molecules.

If the mobility of charge is limited by the availability of suitably sized holes then one diagnostic test for determining where most ionic liquids take on the properties of solutions would be that the molar conductivity passes through a maximum where ion mobility and hole mobility are at equilibrium.

3.5 Poly-Alcohol/Sugar Based Deep Eutectic Solvents

In addition to the di and tri alcohols investigated in sections 3.1., 3.2. and 3.3. other OH containing species were investigated. The polyols investigated are erythritol (14), xylitol (15), D-(-) fructose (16) and D-glucose (17). Each compound was added to ChCl and phase diagrams were constructed for each system.



(14) (15)

(16) (17)

Figure 3.44: *Diagrammatic representations of polyols investigated in section 3.5.*

Table 3.5: *Freezing point data for ChCl:Polyol systems investigated*

Ratio of ChCl:Polyol	**T_f ChCl:Erythritol**	**T_f ChCl:Xylitol**	**T_f ChCl:D-(-) Fructose**	**T_f ChCl:D-Glucose**
2.5:1	N/A	16 ºC	37 ºC	44 ºC
2:1	0 ºC	-4 ºC	10 ºC	15 ºC
1.5:1	-10 ºC	2 ºC	13 ºC	24 ºC
1:1	-16 ºC	-3 ºC	20 ºC	31 ºC
1:1.5	-8 ºC	4 ºC	28 ºC	39 ºC
1:2	30 ºC	7 ºC	33 ºC	N/A
1:2.5	N/A	N/A	36 ºC	N/A
1:3	N/A	N/A	40 ºC	N/A

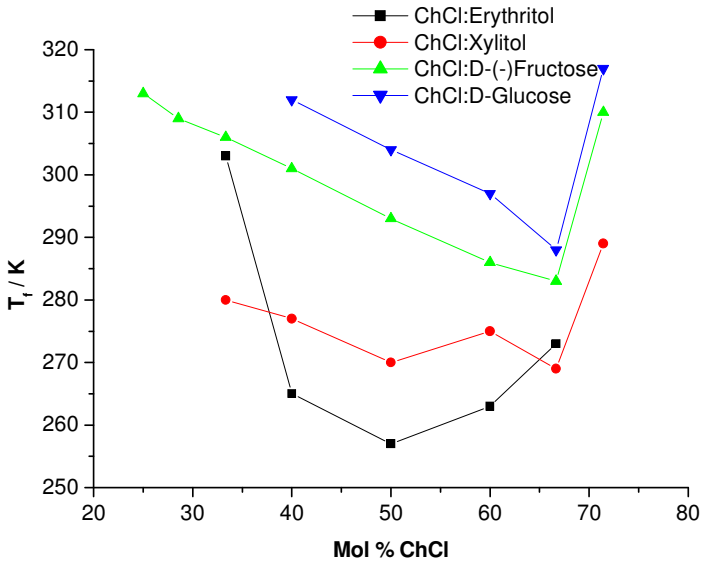


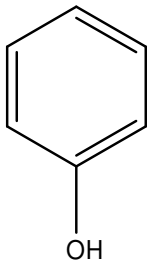
***Figure 3.45**: Phase diagram of ChCl:Polyol mixtures*

From the data shown in Figure 3.45, there are some interesting trends with regards to the number of OH groups there are in the polyol. It can be seen that DES's can be formed in molar ratios where ChCl is in molar excess. The results for the 4 OH containing species erythritol (14), show that the eutectic point occurs at a ratio of 1:1 ChCl:polyol. As the number of OH groups increases to 5, the eutectic point shifts from a ratio of 1:1 to a ratio of 2:1 ChCl:polyol as seen in the D-(-) fructose (16) and D-glucose (17) cases. What is perhaps most interesting about the data shown in Figure 3.44 is that the data for xylitol (15) shows two inflection points suggesting two eutectic points. These point occur at 1:1 which is the same as the erythritol system and at 2:1, the same seen in the D-(-) fructose (16) and D-glucose (17) systems. The erythritol system is thought to form the eutectic at a ratio of 1:1 due to the ChCl being able to hold 4 OH interactions, analogous to the eutectic of diols being at a ratio of 1:2 (2 x 2 OH = 4 OH). Since xylitol, D-(-) fructose and D-glucose all contain 5 OH groups more than one ChCl molecule is thought to interact. Because both D-(-) fructose and D-glucose are in cyclic form the angle of interaction between the Cl anion and OH groups is more favourable to allow 2 ChCl molecules to form hydrogen bonds. This would explain the larger depression in freezing point than in the case of xylitol. Because of the straight chain nature of xylitol it is possible to allow 2 Cl anions to form hydrogen bonds with

OH groups however the repulsion between ChCl molecules is likely to reduce this interaction.

3.6 Mono Functional Alcohol Deep Eutectic Solvents

This chapter has gone into a lot of detail about the use of alcohols with two or more –OH groups to form DES's but it is possible to form a DES from choline chloride and phenol which contains only one –OH group.



(18)

From the viscosity and conductivity data shown in Figures 3.45 and 3.46 respectively, the eutectic point appears to be at a ratio of 4:1 (Phenol:ChCl). The surprising result is perhaps the viscosity of these systems, where the eutectic point exhibits a viscosity of 30 cP at 2 ºC. Due to the extremely low freezing point of some of these mixtures near the eutectic point (< -40 ºC), a phase diagram was not constructed due to the large degree of inaccuracy of measuring the freezing points at such low temperatures.

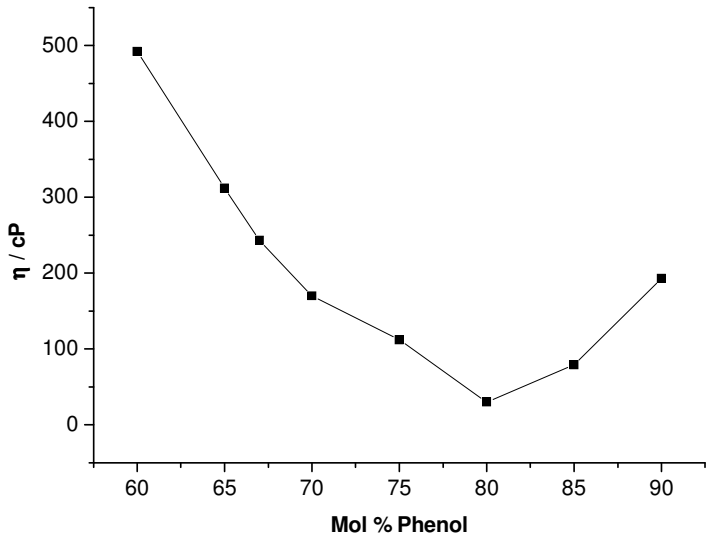


Figure 3.45: Viscosity of ChCl:Phenol mixtures as a function of composition at 275K

The viscosity of this system is amongst one of the least viscous DES's formed from two solid components. Currently it is thought that it offers the lowest viscosity of any ChCl/hydrogen bonding compound mixture. This is attributed to the electron-withdrawing effects of the delocalised π-electron system in the aromatic ring. Through the hydrogen bond between the OH and Cl formed, the aromatic group increases the amount of delocalisation of Cl anion from the choline cation, hence an increase in entropy of the system, which results in the lower viscosity.

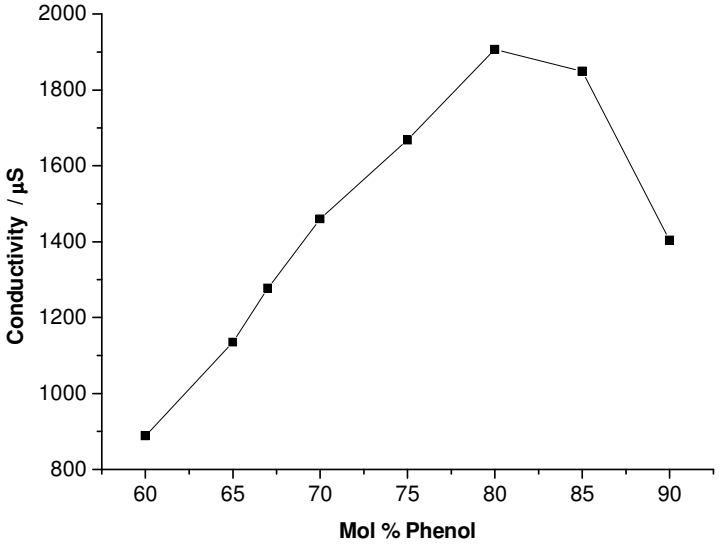


Figure 3.46: Conductivity of ChCl:Phenol mixtures as a function of composition at 275K